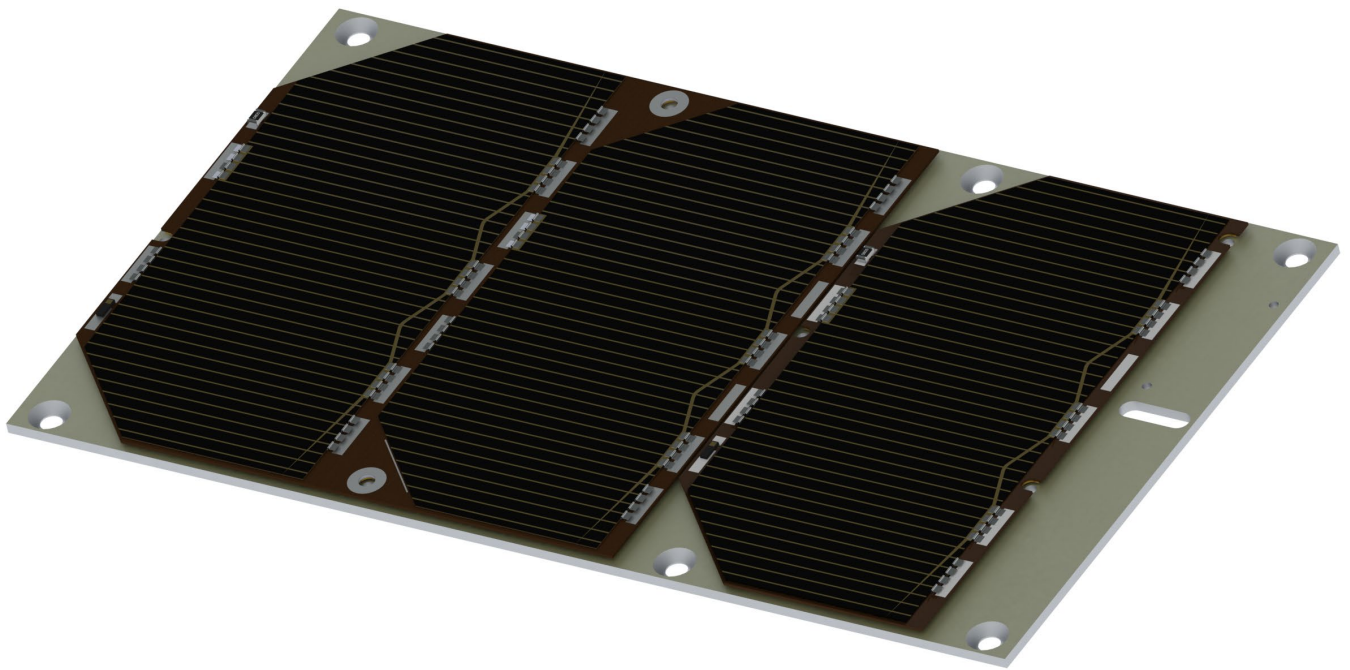




NanoPower MSP

Datasheet

Modular solar panel with integrated coarse sun sensor and temperature sensor.

Product name: NanoPower MSP

Document No.: 1016340

Revision: 2.1

Author: KAME / JBJ

Reviewed by: BGS / FCO

Approved by: LBH

Approval date: 03-02-2021 LBH

Confidentiality Notice

This document is submitted for a specific purpose as agreed in writing and contains information, which is confidential and proprietary. The recipient agrees by accepting this document, that this material will not be used, transferred, reproduced, modified, copied or disclosed in whole or in part, in any manner or to any third party, except own staff to meet the purpose for which it was submitted without prior written consent.

GomSpace © 2021

1 Table of Contents

2	OVERVIEW	4
2.1	Highlighted Features.....	4
3	WARNINGS.....	4
4	DEFINITIONS OF SATELLITE SIDES	4
5	DESCRIPTION OF THE MODULAR SYSTEM	6
5.1	General	6
5.2	External Sensors	6
5.3	Internal Sensors.....	7
5.4	A Side	8
5.4.1	MSP A-6-1	9
5.4.2	MSP A-5-1	9
5.4.3	MSP A-4-1	10
5.4.4	MSP A-3-1	10
5.4.5	MSP A-2-1	11
5.4.6	MSP A-1-1	11
5.5	B Side	12
5.5.1	Central covering plates	13
5.6	C and D Side	13
6	MODULE DESCRIPTIONS.....	14
6.1	Solar cell (Dual)	14
6.1.1	Anodecell	14
6.1.2	Dualcell	15
6.2	Gomspace Sensor System Bus.....	15
6.2.1	Connector Location.....	15
6.2.2	J1 – Fine Sun Sensor	16
6.2.3	J2 – GSSB IN	16
6.2.4	J3 – GSSB OUT	16
6.2.5	J4 – GomSpace	17
6.3	Connect to solar cells.	17
6.3.1	Connector Location, S2C.....	17
6.3.2	J1 – Serial Connector 1	17
6.3.3	J2 – EPS Connector 2	18
6.3.4	Connector Location, P2C.....	18
6.3.5	J1 – Serial Connector 1	18
6.3.6	J2 – EPS Connector 2	18
7	ABSOLUTE MAXIMUM RATINGS.....	19
8	ELECTRICAL CHARACTERISTICS	19
9	PHYSICAL CHARACTERISTICS.....	20
9.1	Covering Plates	20
10	DISCLAIMER	20

2 Overview

The NanoPower MSP is a modular designed solar panel system mainly designed to be used on Gomspace satellites. The 2 different cell PCBs can be combined in many different combinations covering a wide range of satellites; also satellites not designed by Gomspace. The unique modular design ensures that besides antennas and sensors the satellite can be fully covered with solar panels to optimize energy production.

The MSP system is designed to fit on the GomSpace 6U structure, however they also fit on structures from other vendors – for more information contact GomSpace.

The MSP is designed to fit with several other GomSpace products:

- NanoCom ANT-6F
- NanoStructure 6U
- NanoUtil AR6
- NanoUtil MSP-FPP

2.1 Highlighted Features

- 30% solar panel efficiency
- Up to 1.2 W per cell in LEO
- AzurSpace 3G30A space qualified triple junction solar cell assemblies
- CMG 100 cover glass 100 μm
- 30.18 cm^2 effective solar cell area.
- Coarse sun sensor
- External and internal temperature sensors
- Internal temperature -40°C to $+85^{\circ}\text{C}$
- External temperature -55°C to $+175^{\circ}\text{C}$

3 Warnings



Handling

This product uses advanced solar cells that are fragile. Do not touch solar cells.

**Only handle solar panels without touching solar cells
or their tabs**

Never place anything on solar cells!

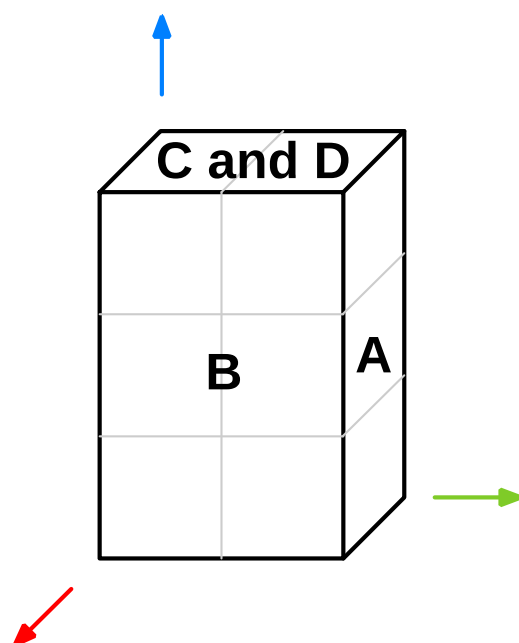


ESD

This product uses semiconductors that can be damaged by electrostatic discharge (ESD). When handling, care must be taken so that the devices are not damaged. Use appropriate precautions.

4 Definitions of Satellite Sides

The different sides of the satellite are referred to as illustrated in the drawing below.



The satellite surface can be covered with solar cells to collect as much energy as possible. The payload in the satellite traditionally require as much energy as possible to achieve higher duty cycles.

The naming of modules is referring to the satellite side for mounting.

MSP **A**-x-x means that the module is developed with the A side for mounting, primary. Small MSP A modules can also in special situations be placed on side C/D.

The naming of the modules is also informing about the module's connection.
Solae cells can be combined in many different combinations, serial or parallel and combinations of these.
Gomspace have decided to have information about combination included in the module name.

MSP A-6-1: 6 solar cells are combined into 1 output.

MSP A-3-1: 3 solar cells are combined into 1 output.

MSP A-4-4: 4 times 4 solar cells are combined into 4 output.

Other combinations upon request.

5 Description of the Modular System

5.1 General

The A, C and D modules is built up on basis of an aluminium plate consists; 1.5 mm aluminium plate. The B side thickness is 3 mm due to the large size. On the B side, the weight is reduced by removing material where possible to a thickness of 1.5 mm.

Each module has one temperature sensor and one coarse sun sensor. The backside holds the connections to ESP (S2C and P2C) and MSP-GSSB module.

The amount of energy that can be harvested from the modules is up to 1.2W for each cell, when directly facing the Sun. **Up to** has to be taken into account because the angle to the sun is very important, and mis-angling is directly measurable in reduced energy harvesting.

5.2 External Sensors

Next to the top solar cell on each unit, there is a temperature sensor and a coarse sun sensor. The sensors are connected to the GSSB PCB on the back, see more in chapter 6. View the illustration below.

Coarse sun sensor on the left and the temperature sensor on the right.

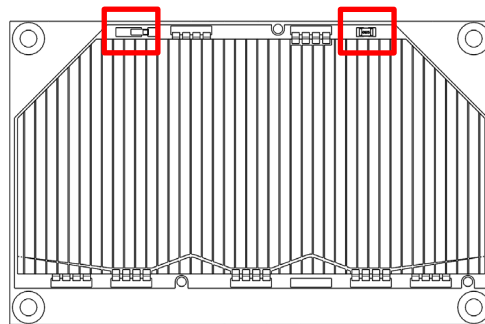
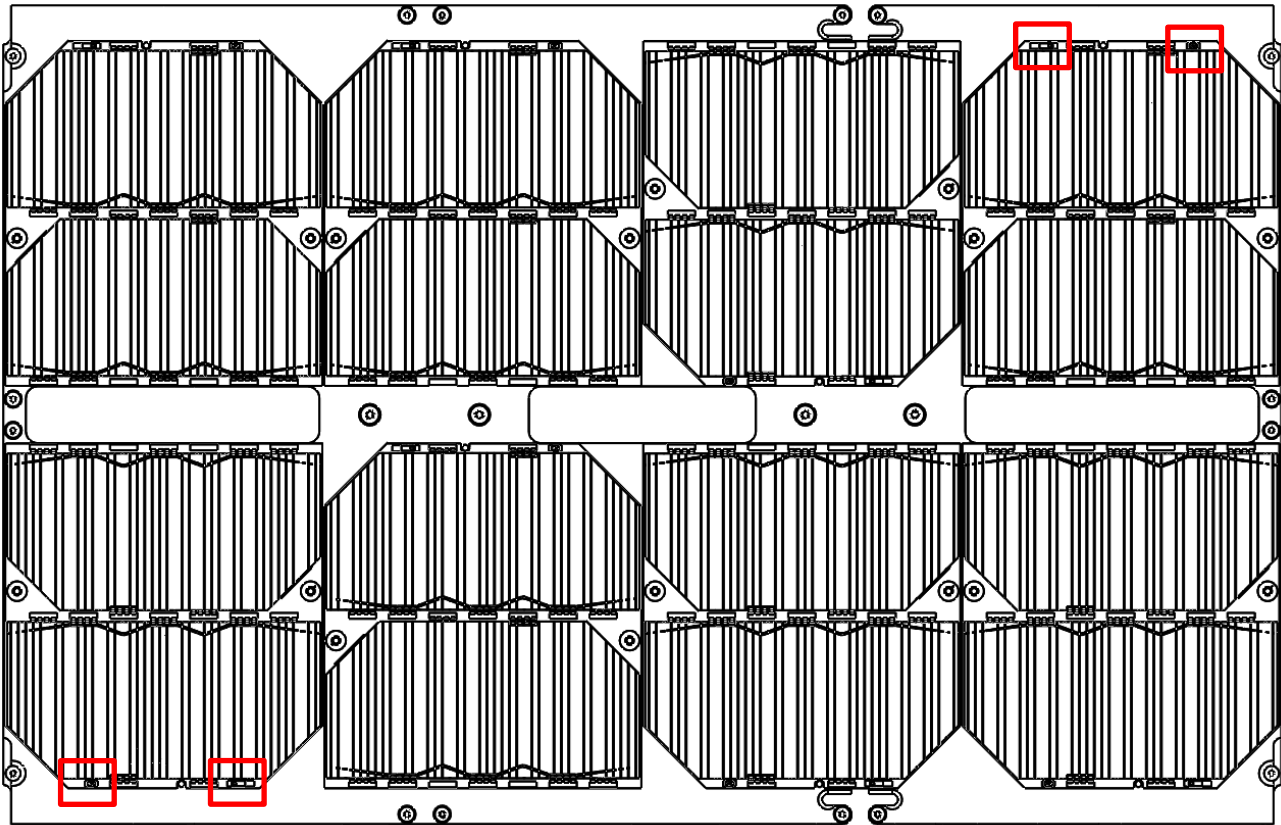


Figure 1 AnodeCell

On the B side, there is a pair in the bottom left and top right corner for the MSP-B-4-4 version. The other MSP-B-X-X versions will have at least one temperature sensor and a coarse sun sensor.



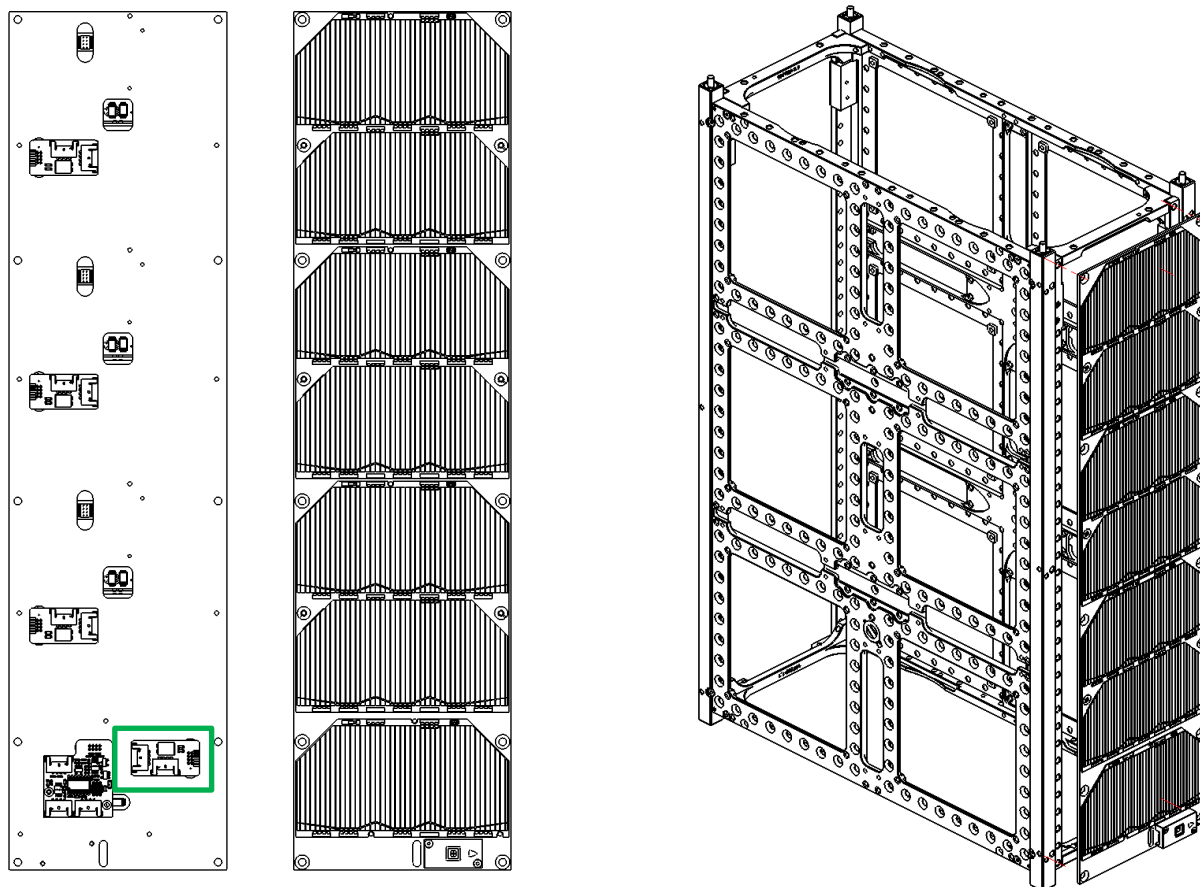
5.3 Internal Sensors

The internal temperature sensor is in the MCU on the GSSB module. View chapter 6.

5.4 A Side

An A side plate module can contain from one to seven solar cells. Each module has a connector to the EPS and a connector to GSSB. The solar panels are connected in one serial string in each module. The S2C to the EPS is shown in green in each drawing where there is more than one S2C.

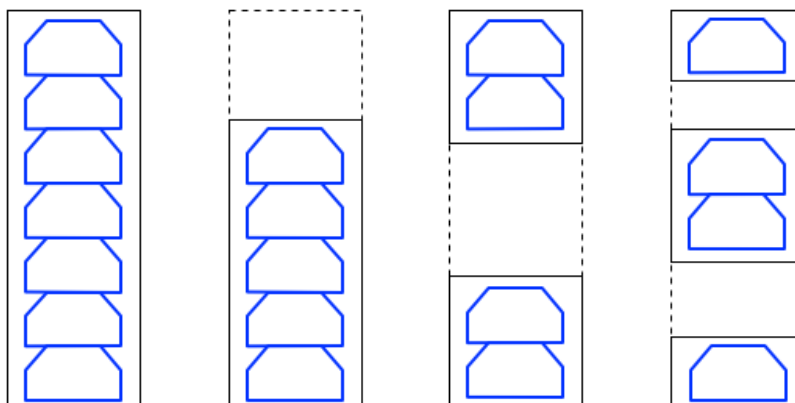
Below is shown a drawing of a 7-cell unit, front and back. And a drawing to illustrate mounting on a 6U structure.



On each module, mounting holes are placed at the top and bottom, and otherwise where it can fit the NanoStructure 6U holes.

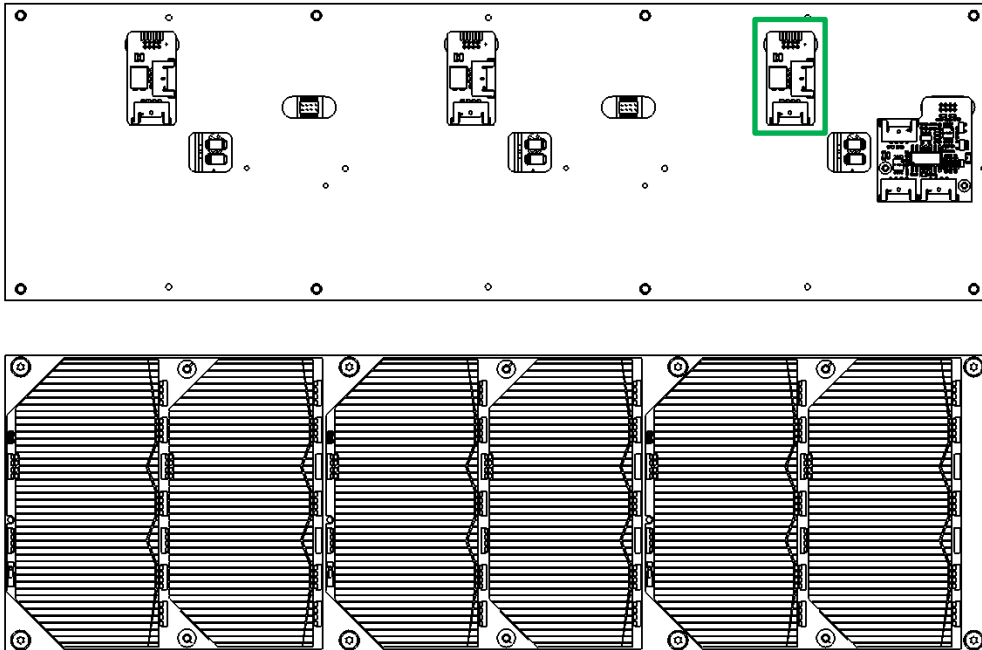
Several smaller modules can be fitted on one side, e.g. if an antenna is required to be placed in the middle area then modules can be placed above and below it.

Examples of setups:



5.4.1 MSP A-6-1

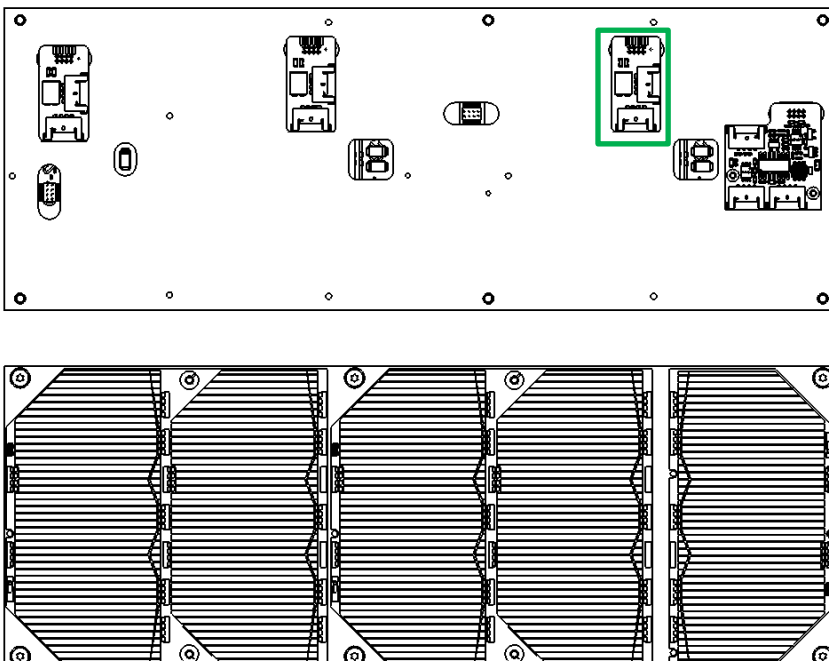
Here the front and back side of the MSP-A-6-1 is shown.



Output in best case is 14.4V @500mA = 7.2W

5.4.2 MSP A-5-1

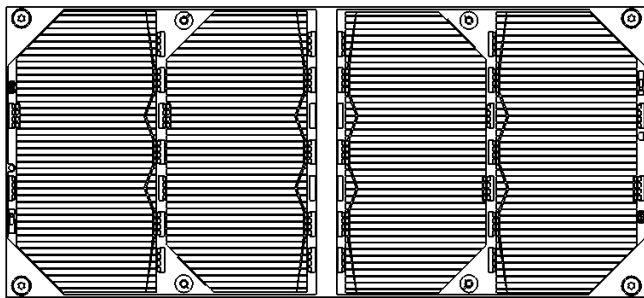
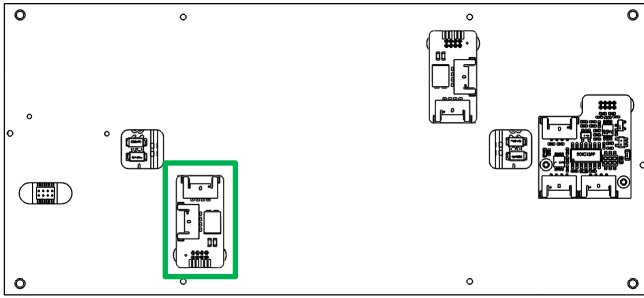
Here the front and back side of the MSP-A-5-1 is shown.



Output in best case is 12V @500mA = 6.0W

5.4.3 MSP A-4-1

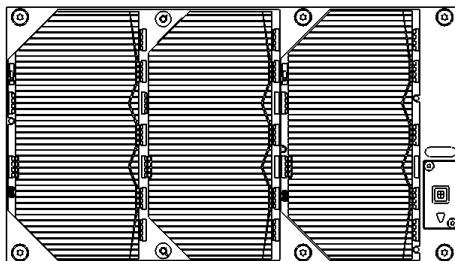
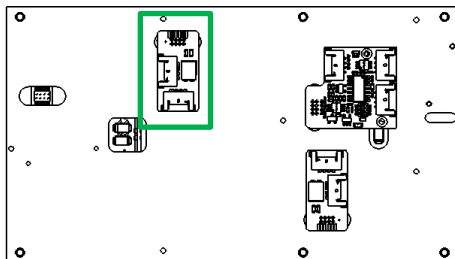
Here the front and back side of the MSP-A-4-1 is shown.



Output in best case is 9.6V @500mA = 4.8W

5.4.4 MSP A-3-1

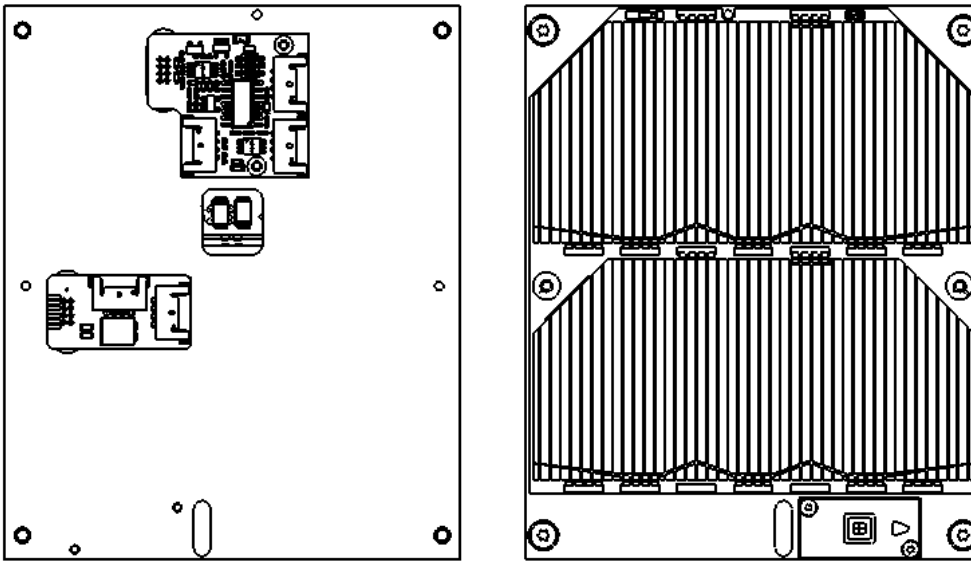
Here the front and back side of the MSP-A-3-1 is shown.



Output in best case is 7.2V @500mA = 3.6W

5.4.5 MSP A-2-1

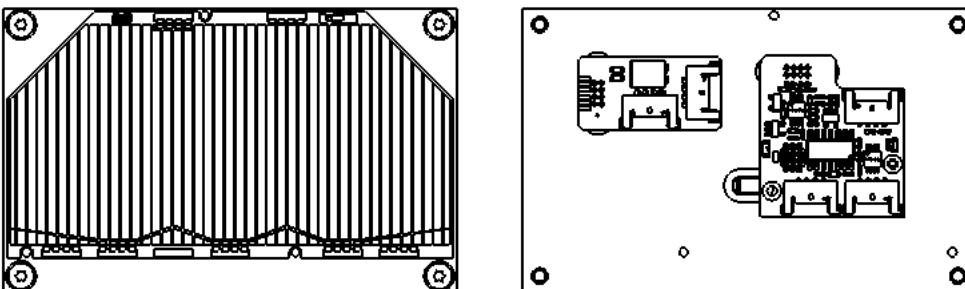
Here the front and back side of the MSP-A-2-1 is shown.



Output in best case is $4.8V @ 500mA = 2.4W$

5.4.6 MSP A-1-1

Here the front and back side of the MSP-A-1-1 is shown.

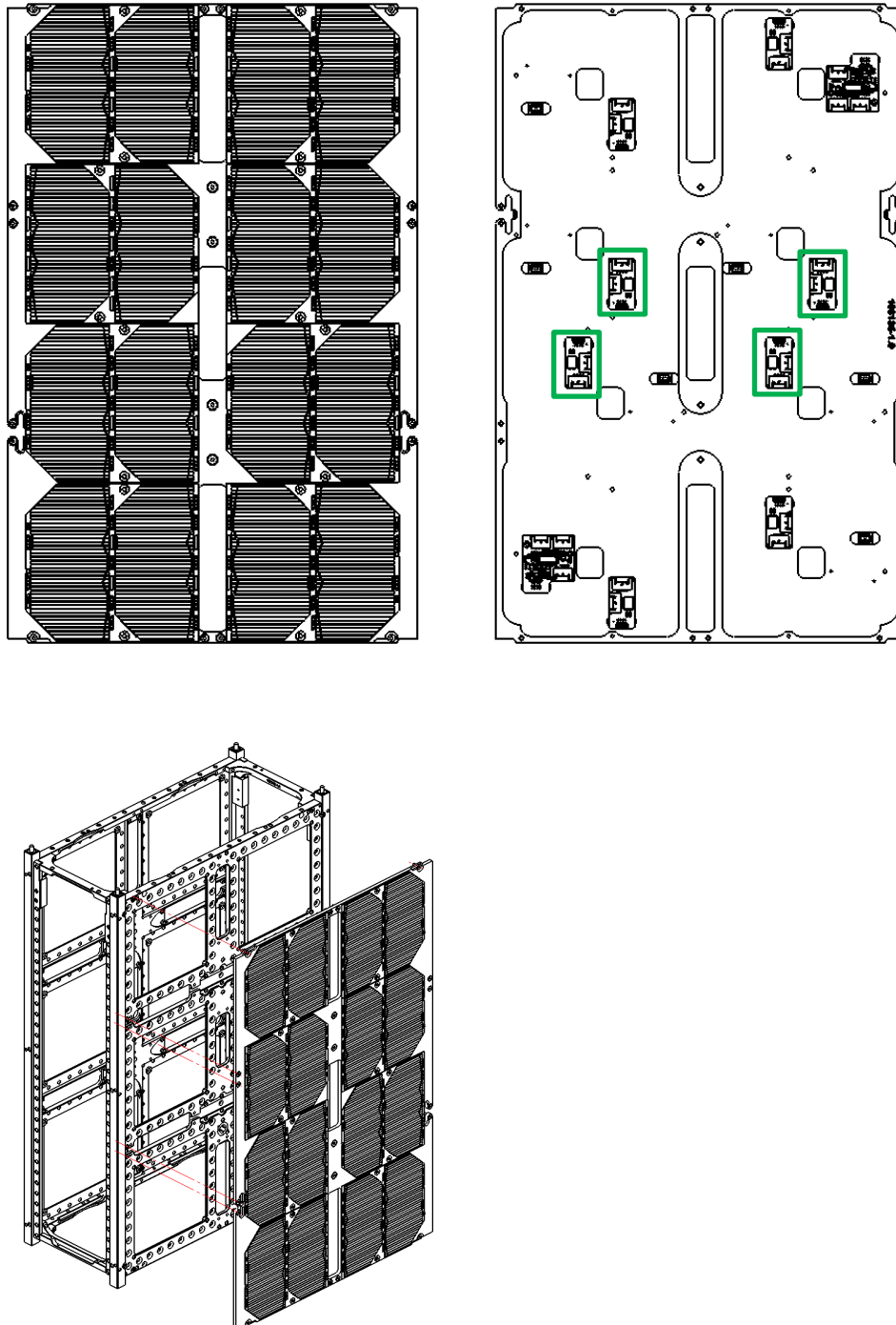


Output in best case is $2.4V @ 500mA = 1.2W$

5.5 B Side

The B side is one plate covering the entire side, mounted with 16 solar panels. Down the middle are three cut outs for external sensors, such as Fine Sun Sensors. The solar panels can be connected as either a 4 serial strings / 4 parallel or an 8 serial / 2 parallel string.

Below is shown the front and back. And a drawing to illustrate mounting on a 6U structure.



The B-8-2 panel is built as the B-4-4 and configured to B-8-2 with the harness.

MSP B-4-4 output in best case is $9.6\text{V} @ 500\text{mA} = 4.8\text{W} * 4 = 19.2\text{W}$

MSP B-8-2 output in best case is $19.6\text{V} @ 500\text{mA} = 9.6\text{W} * 2 = 19.2\text{W}$

5.5.1 Central covering plates

Down the middle of the B side there are three rectangular cut out. Two kinds of covering plates are made to fit here:

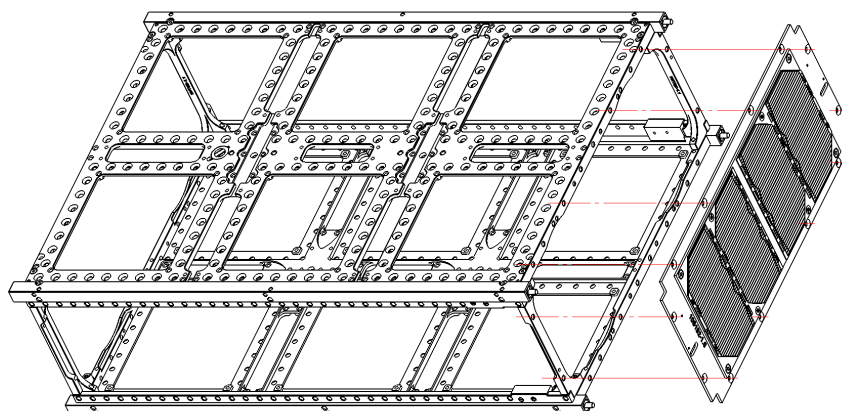
- 1.5 mm aluminium covering plate
- 1.5 mm aluminium plate with a sunk slot where a GomSpace NanoSense Fine Sun Sensor (FSS) can fit.



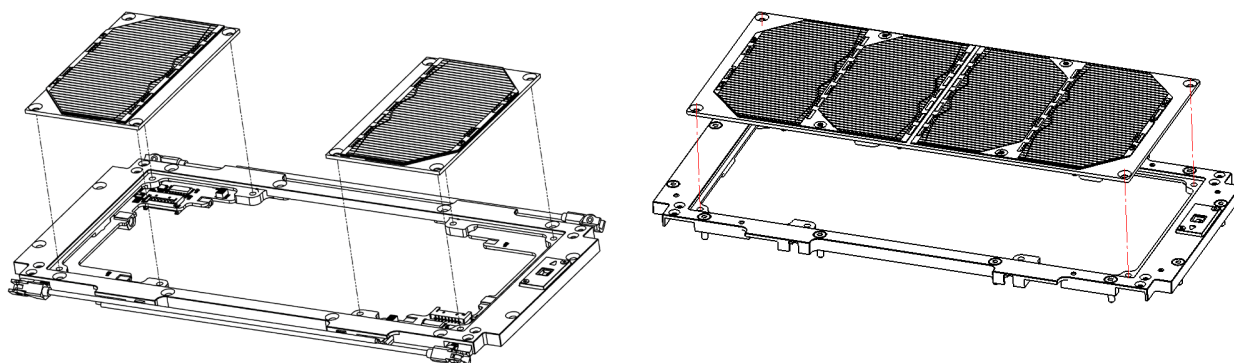
A set of three of each plate is delivered with the B side.

5.6 C and D Side

The C-side is a bracket mounted with a MSP-A-4-1, and the MSP-A-4-1 is shown in section 5.4.3.



The D-side is if the satellite is equipped with a GomSpace NanoCom ANT-6F. The ordering of the module is done with the ANT-6F and optionsheet. It can be mounted with either two modules with one solar cell each, or with one module with four solar cells. The ANT-6F mounting options is illustrated in the figure below.



6 Module descriptions

6.1 Solar cell (Dual)

The main components in the MSP is the solar cells, components transforming incoming sunlight into electric energy. In the market different suppliers are providing solar cells, here we are concentrating in a few vendors, these vendors are all delivering highly efficient solar cells to be used in space.

Gomspace has developed some surrounding components and thereby been able to provide solar cell modules to the market.

Gomspace are providing 2 type of modules; Anodecell and Dualcells.

6.1.1 Anodecell

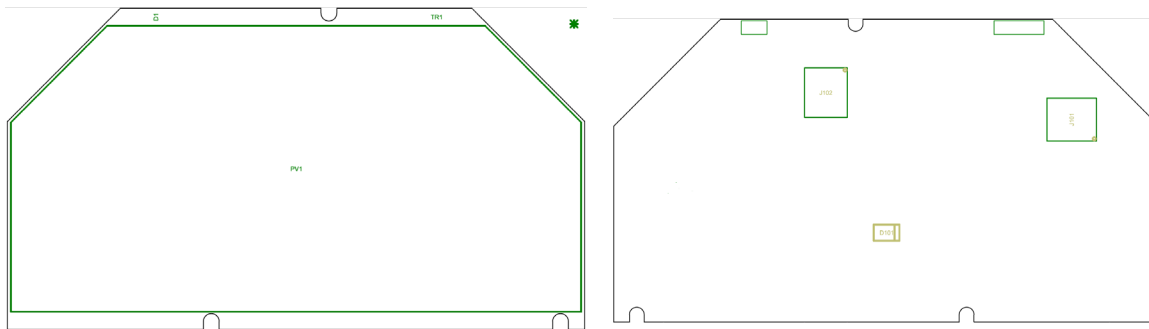


Figure 2 Anodecell

The Anodecell consist of only one solar cell, one light sensor and one temperature sensor; connections to MSP-GSSB and P2C/S2C is placed on backside. The module is 81 times 45 mm.

6.1.2 Dualcell

The Dualcell consist of two solar cells, one light sensor and one temperature sensor; connections to MSP-GSSB and P2C/S2C is placed on backside. The module is 81 times 90 mm.

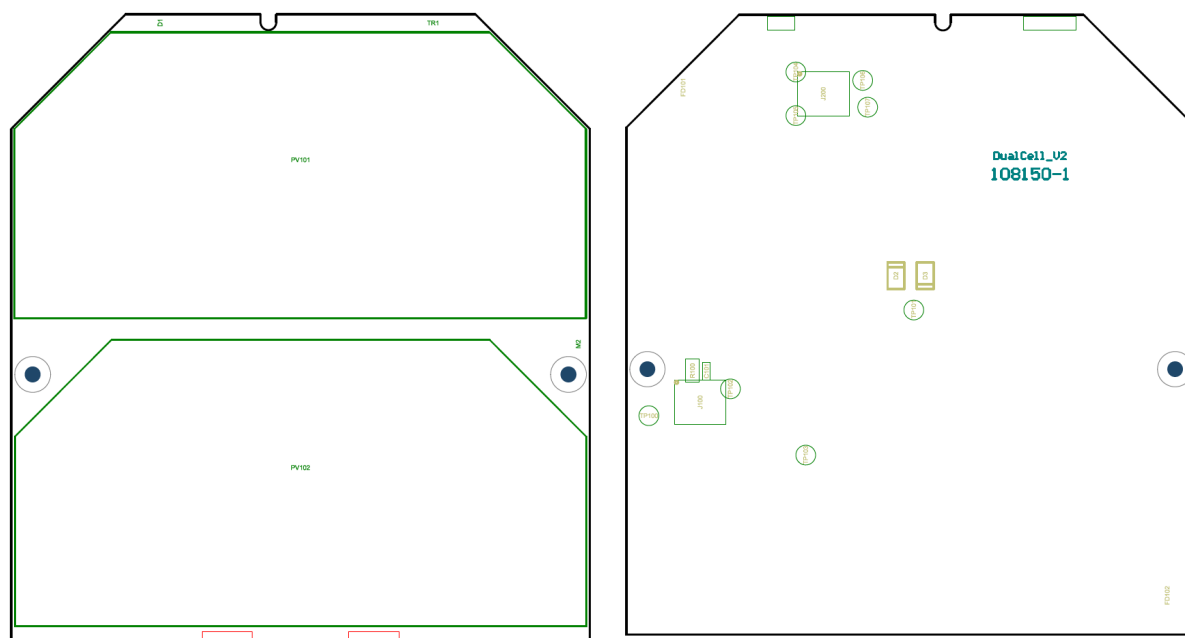


Figure 3 Dualcell

6.2 Gomspace Sensor System Bus

The MSP-GSSB modules is to be placed on the solar array backside, connected to the solar array via a soldered connection. The MSP-GSSB module interface the sun sensor and temperature sensor to the system bus by an onboard processor. The processor will readout the values from temperature and sun sensor and transmit these values to the main system via I2C connection.

FSS; Fine Sun Sensor is not further explained in this document.

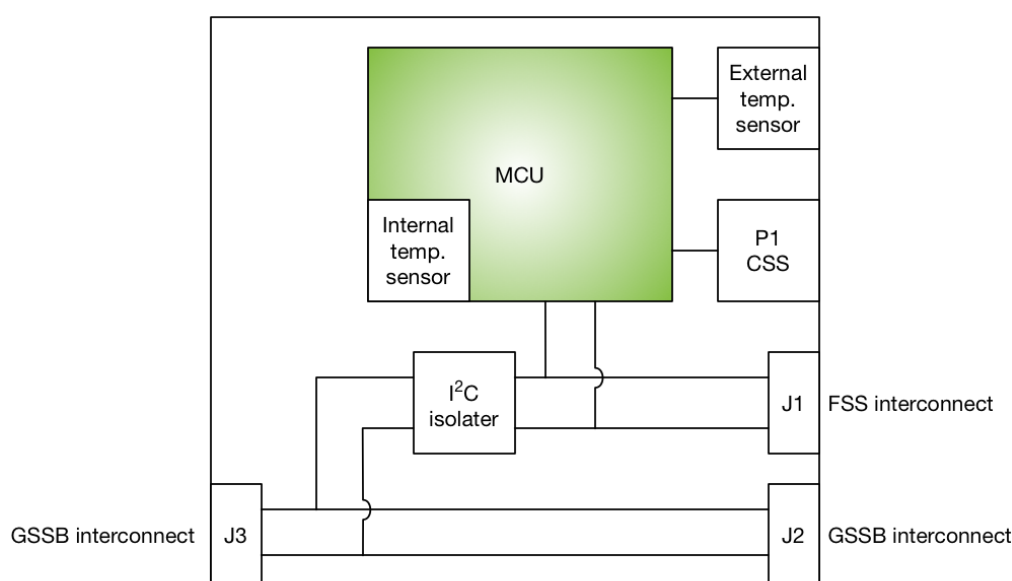


Figure 4 MSP-GSSB block diagram

6.2.1 Connector Location

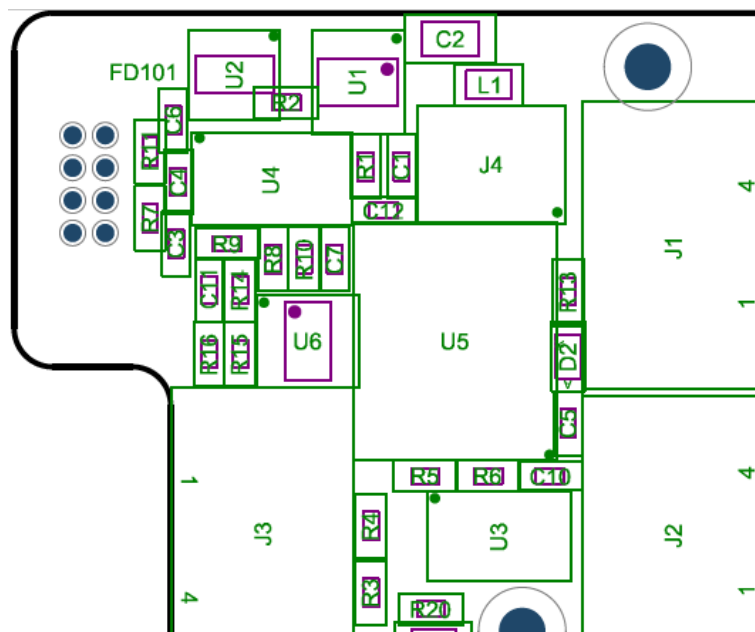


Figure 5 MSB-GSSB component placement

On the top left are the interface connector to the front sensors.

6.2.2 J1 – Fine Sun Sensor

Molex Pico-Lock 1.50 mm pitch, right angle, 504050-0491.

Pin	Description
1	GSSB_VCC
2	GND
3	SDA
4	SCL

6.2.3 J2 – GSSB IN

Molex Pico-Lock 1.50 mm pitch, right angle, 504050-0491.

Pin	Description
1	GND
2	GSSB_SCL
3	GSSB_SDA
4	GSSB_VCC

6.2.4 J3 – GSSB OUT

Molex Pico-Lock 1.50 mm pitch, right angle, 504050-0491.

Pin	Description
1	GSSB_VCC
2	GSSB_SDA
3	GSSB_SCL
4	GND

The pins on connector J2 and J3 are connected back to back. This is to avoid split harness and the sensor bus can be routed through the GSSB module.

6.2.5 J4 – GomSpace

Samtec CLP-103-02-G-D-BE

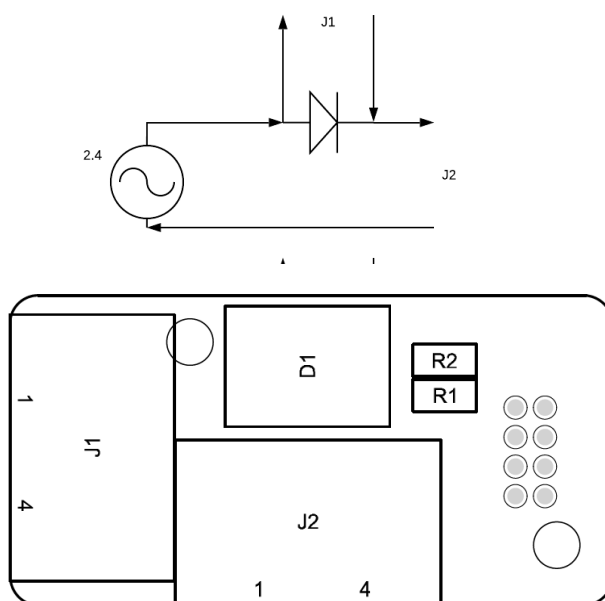
This connector is used for programming the processor on the PCB.

6.3 Connect to solar cells.

On the system we can connect the solar cells in different combinations; serial or parallel. If solar cells are located to point in same direction it can be convenient to connect cells in either parallel or serial; mainly to simplify the harness but also because we do not have unlimited numbers of connections into the power system. Here we have the module for connection the solar cells in serial; S2C.

By connecting more S2C boards correct we can connect 2 or more solar modules in serial, by connecting solar cells in serial we raise the voltage but keep the current low.

6.3.1 Connector Location, S2C.



The diode is implemented to have a reverse voltage protection. If more strings are put into parallel connection, for example in the harness, it can occur that one series of strings can load another set of strings.

6.3.2 J1 – Serial Connector 1

Molex Pico-Lock 1.50 mm pitch, right angle, 504050-0491.

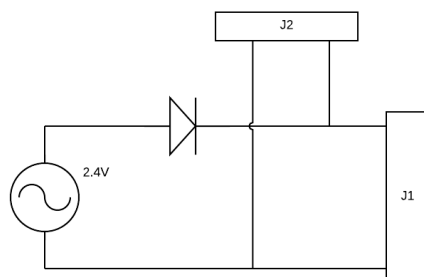
Pin	Description
1	Cathode
2	Cathode
3	Anode
4	Anode

6.3.3 J2 – EPS Connector 2

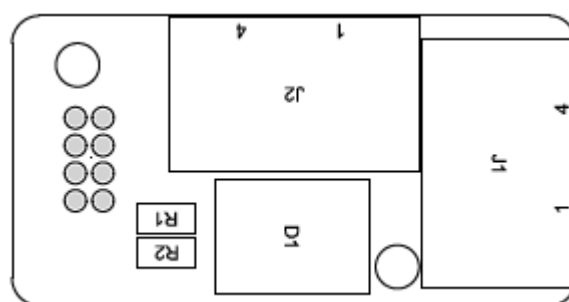
Molex Pico-Lock 1.50 mm pitch, right angle, 504050-0491.

Pin	Description
1	Cathode
2	Cathode
3	Anode (blocking diode)
4	Anode (blocking diode)

6.3.4 Connector Location, P2C.



P2C Schematic



P2C component placement.

The diode is implemented to have a reverse voltage protection. If more strings are put into parallel connection, for example in the harness, it cannot occur that one series of strings can load another set of strings, due to the diode.

6.3.5 J1 – Serial Connector 1

Molex Pico-Lock 1.50 mm pitch, right angle, 504050-0491.

Pin	Description
1	Cathode
2	Cathode
3	Anode
4	Anode

6.3.6 J2 – EPS Connector 2

Molex Pico-Lock 1.50 mm pitch, right angle, 504050-0491.

Pin	Description
1	Cathode
2	Cathode
3	Anode (blocking diode)
4	Anode (blocking diode)

7 Absolute Maximum Ratings

Stresses above those listed under Absolute Maximum Ratings may cause permanent damage to the MSP. Exposure to absolute maximum rating conditions for extended periods may affect the reliability.

Symbol	Description	Min.	Max.	Unit
T_{ext}	Temperature on the solar panel	-55	+175	°C
T_{int}	Electronics operating temperature	-40	+85	°C
GSSB_VCC		3.1	3.5	V

8 Electrical Characteristics

Parameter	Condition	Min	Typ.	Max	Unit
Single Solar Panel	Full sunlight in LEO				
• Voltage	Optimal voltage	2.32		2.42	V
• Current	Current at optimal voltage	490		508	mA
• Power	Maximum power	1135		1200	mW
• Efficiency		29.8	30	30.2	%
GSSB_VCC			3.3		V
GSSB_I_idle			2.6		mA
Course Sun Sensor	Short current at 1367 W/m ²		930		μA
• Current			1.85	3.5	°
• Cosine error					
Temperature Sensor External		-75		+175	°C
• Range				3.5	%
• Resolution					
Temperature Sensor Internal		-40		85	°C
• Range		-10		10	%
• Resolution					

9 Physical Characteristics

Module plates are made of AL5005-H14 (clear anodization)

Values are of modules with filled solar panels

Name	Configuration	Plate Size [mm]	Mass
MSP-A-1-1	Single	51.7 x 82.6 x 1.5	32
MSP-A-2-1	2 in series	100.5 x 82.6 x 1.5	60
MSP-A-3-1	3 in series	146.3 x 82.6 x 1.5	81
MSP-A-4-1	4 in series	182.9 x 82.6 x 1.5	103
MSP-A-5-1	5 in series	228.7 x 82.6 x 1.5	135
MSP-A-6-1	6 in series	274.5 x 82.6 x 1.5	162
MSP-A-7-1*	7 in series	326.5 x 82.6 x 1.5	190
MSP-B-4-4	4 serial, 4 parallel	326.5 x 209.0 x 3	453
MSP-B-8-2	8 serial 2 parallel	326.5 x 209.0 x 3	453
MSP-B-4-1	4 serial	114.9 x 209 x 3	145
MSP-C-4-1	4 in series	221.5 x 97.5 x 1.5	132
MSP-D-4-1	4 in series	189.0 x 83.6 x 1.5	112

* Note that if the seven solar cell version is chosen, a few mm of extra aluminium plate is added, so the plate covers the entire side.

9.1 Covering Plates

The covering plates with FSS mounting are made of AL5005 and has a chromital TCP surface treatment. The other covering plates are made of AL5005-H14 clear anodization.

Description	Value	Unit
Mass (short)	4.4	g
Mass (long)	5.3	g
Mass (long with FSS mounting)	4.7	g
Size (short)	57.0 x 14.0 x 1.5	mm
Size (long)	74.0 x 14.0 x 1.5	mm
Size (long with FSS mounting)	74.0 x 14.0 x 5.5 (1.5)	mm

10 Disclaimer

The information in this document is subject to change without notice and should not be construed as a commitment by GomSpace. GomSpace assumes no responsibility for any errors that may appear in this document.

In no event shall GomSpace be liable for incidental or consequential damages arising from use of this document or the software and hardware described in this document.